

COMMON TERMS & SYMBOLS – Lectures 5-8

σ : **Population standard deviation (sigma)**. If known, you can use the Normal distribution for confidence intervals and hypothesis tests. However, this is often unknown, and must be estimated with s .

s : **Sample standard deviation**. If σ is unknown, this will estimate it, but you must use the t-distribution for confidence intervals and hypothesis tests to account for this estimation. Note that you can always use the t-distribution, but if σ is known, the Normal distribution will be more effective.

The **sd()** function will give the sample standard deviation for a vector in an imported data set.

μ & p : **Population parameter**. These are the population mean and population proportion, respectively. This is what you want to estimate or make conclusions about using your sample statistic.

$\bar{X}$ & $\hat{p}$: **Sample statistic**. These are your estimates for the population mean and population proportion, respectively. Which one you use depends which parameter you are estimating. If you have a value for μ or p from a null hypothesis, a hypothesis test measures how far away (in standardized terms) from that value the sample mean is.

The **mean()** function will give the sample mean for a vector in an imported data set.

z & t : **Test statistics** for the **Normal** and **t** distributions, respectively. They give you a standardized measure of how “unusual” your sample statistic is in relation to a null hypothesis value.

Test statistics take a common form: (sample statistic – H_0 value)/(error term)

α : **Confidence level**. This is the largest probability of rejecting H_0 when it is actually true. Typically seen in values no greater than 0.05—lower α reduced the chance of false positives, and requires a more extreme test statistic to reject H_0 .

p-value: The probability of observing the test statistic you do, or something **more extreme**, under the null hypothesis H_0 . A low p-value (less than **α**) suggests your null hypothesis should be rejected, as your test statistic is unlikely to occur under it.

pnorm() and **pt()** are function used to calculate p-value. Note that the calculation itself varies depending on whether you perform a one- or two-sided test.

Example: $2*(1-pnorm(|z|))$ gives the p-value for a two-sided z-test.

$\sigma/\sqrt{n}$: **Standard error**. This is the error term and denominator for z and one part of the margin of error when σ is known and thus you’re using the standard Normal distribution.

$s/\sqrt{n}$: **Sample error**. This is the error term and denominator for t and one part of the margin of error when σ is unknown and thus you’re using the t-distribution. You can get this from an imported data set with the **sd()** function applied to the variable in question.